

Conjugate Heat Transfer - Manufactured Solution

May 2026

Manufactured solution

This report defines a steady MMS pair on $\Omega = [0, 1] \times [0, 1]$.

Exact fields

$$u_x(x, y) = \sin(\pi x) \cos(\pi y)$$

$$u_y(x, y) = -\sin(\pi y) \cos(\pi x)$$

$$p(x, y) = \sin(\pi y) \cos(\pi x)$$

$$T(x, y) = \sin(\pi y) \cos(2\pi x)$$

Governing equations

For momentum we assume the steady incompressible Navier–Stokes residual $(u \cdot \nabla)u - \nu \Delta u + \nabla p - f = 0$, with $\nabla \cdot u = 0$. For temperature we assume steady advection–diffusion $u \cdot \nabla T - \kappa \Delta T - s = 0$, where κ is thermal diffusivity.

Source terms

$$f_x(x, y) = \pi (2\pi\nu \cos(\pi y) - \sin(\pi y) + \cos(\pi x)) \sin(\pi x)$$

$$f_y(x, y) = \frac{\pi (2\pi\nu \sin(\pi(x-y)) - 2\pi\nu \sin(\pi(x+y)) + \sin(2\pi y) + \cos(\pi(x-y)) + \cos(\pi(x+y)))}{2}$$

$$\nabla \cdot u = 0$$

$$s(x, y) = \frac{\pi (-20\pi\kappa \sin^2(\pi x) + 10\pi\kappa - 3 \cos(\pi x) \cos(\pi y) + \cos(3\pi x) \cos(\pi y)) \sin(\pi y)}{2}$$

Boundary values on $\partial\Omega$

u_x

$$x = 0 : 0$$

$$x = 1 : 0$$

$$y = 0 : \sin(\pi x)$$

$$y = 1 : -\sin(\pi x)$$

u_y

$$x = 0 : -\sin(\pi y)$$

$$x = 1 : \sin(\pi y)$$

$$y = 0 : 0$$

$$y = 1 : 0$$

p

$$x = 0 : \sin(\pi y)$$

$$x = 1 : -\sin(\pi y)$$

$$y = 0 : 0$$

$$y = 1 : 0$$

T

$$x = 0 : \sin(\pi y)$$

$$x = 1 : \sin(\pi y)$$

$$y = 0 : 0$$

$$y = 1 : 0$$